

Genomic Database



Explore Data

What is NCBI?

WWW

.Dot **ncbi**

National Center for Biotechnology Information

.Dot **nlm**

National Library of Medicine

.Dot **nih**

National Institutes of Health



Main goals of consortium

Since 1988

- Establish public databases
- Research in computational biology
- Develop software tools for data analysis
- Disseminate biomedical information

Important NCBI Resources / Databases

www.ncbi.nlm.nih.gov/nucleotide

www.ncbi.nlm.nih.gov/protein

www.ncbi.nlm.nih.gov/pubmed

Sequences

- Gene
- Nucleotide
- Protein
- Assembly

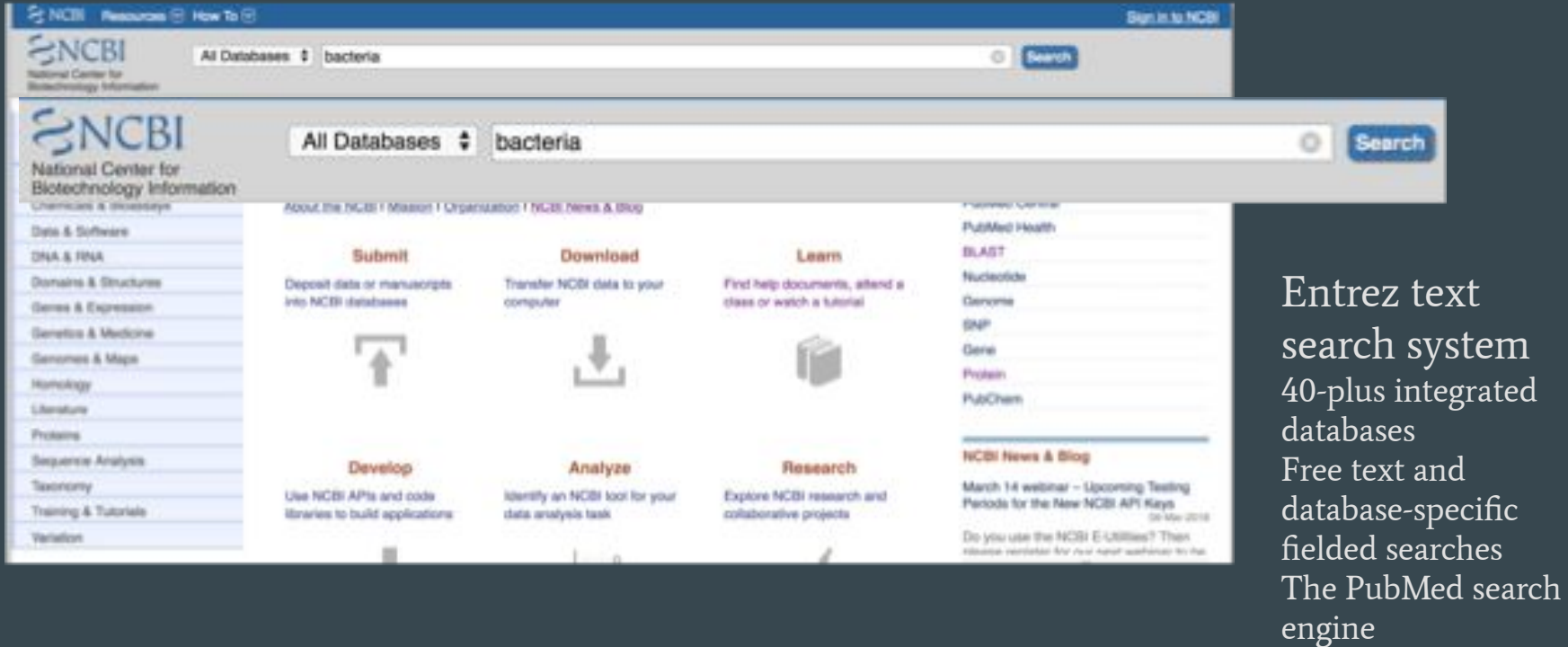
Structures & Chemicals

- Structure
- Conserved Domains
- PubChem

Literature

- PubMed
- PubMed Central (PMC)

NCBI Search Systems



The screenshot displays the NCBI (National Center for Biotechnology Information) homepage. At the top, there is a navigation bar with links for 'NCBI', 'Resources', and 'How To'. Below this, a search bar is visible with the text 'All Databases' and 'bacteria' entered, followed by a 'Search' button. The main content area features a large search bar with the same text and a 'Search' button. To the left of the main content area is a sidebar with a list of search categories: Data & Software, DNA & RNA, Domains & Structures, Genes & Expression, Genetics & Medicine, Genomes & Maps, Homology, Literature, Proteins, Sequence Analysis, Taxonomy, Training & Tutorials, and Variation. The main content area is divided into six sections: Submit (Deposit data or manuscripts into NCBI databases), Download (Transfer NCBI data to your computer), Learn (Find help documents, attend a class or watch a tutorial), Develop (Use NCBI APIs and code libraries to build applications), Analyze (Identify an NCBI tool for your data analysis task), and Research (Explore NCBI research and collaborative projects). On the right side of the main content area, there is a list of featured content: PubMed Central, PubMed Health, BLAST, Nucleotide, Genome, SNP, Gene, Protein, and PubChem. Below this list is a section for 'NCBI News & Blog' with a link to a March 14 webinar and a link to a webinar for the New NCBI API Keys.

Entrez text
search system
40-plus integrated
databases
Free text and
database-specific
fielded searches
The PubMed search
engine

NCBI Search Systems

The screenshot shows the 'Standard Nucleotide BLAST' web interface. At the top, there are tabs for 'blastn', 'blastp', 'blastx', 'tblastn', and 'tblastx'. Below the tabs, a text box labeled 'Enter Query Sequence' contains a FASTA sequence starting with '>gnl|SRA|SRR5483149.1.1'. To the right of the text box is a 'Clear' button and a 'Query subrange' section with 'From' and 'To' input fields. Below the text box is a section for 'Or, upload file' with a 'Choose File' button and 'No file chosen' text. Below that is a 'Job Title' field containing 'gnl|SRA|SRR5483149.1.1'. To the right of the job title is a text box for 'Enter a descriptive title for your BLAST search'. Below the job title is a checkbox labeled 'Align two or more sequences'. Below the checkbox is a 'Choose Search Set' dropdown menu. The dropdown menu is open, showing a list of search sets. The first two options are 'Human genomic plus transcript (Human G+T)' and 'Mouse genomic plus transcript (Mouse G+T)'. The third option is '16S ribosomal RNA sequences (Bacteria and Archaea)', which is highlighted with a blue checkmark. Below the dropdown menu is a 'Database' section with a list of databases. The first two options are 'Reference RNA sequences (refseq_rna)' and 'RefSeq Representative genomes (refseq_representative_genomes)'. The third option is 'RefSeq Genome Database (refseq_genomes)', which is highlighted with a blue checkmark. Below the database list is an 'Organism' section with a list of organisms. The first two options are 'Whole-genome shotgun contigs (wgs)' and 'Expressed sequence tags (est)'. The third option is 'Sequence Read Archive (SRA)', which is highlighted with a blue checkmark. Below the organism list is a 'Limit to' section with a list of limits. The first two options are 'Transcriptome Shotgun Assembly (TSA)' and 'High throughput genomic sequences (HTGS)'. The third option is 'Patent sequences (pat)', which is highlighted with a blue checkmark. Below the limit list is an 'Entrez Query' section with a list of queries. The first two options are 'Protein Data Bank (pdb)' and 'Reference genomic sequences (refseq_genomic)'. The third option is 'Human RefSeqGene sequences (RefSeq_Gene)', which is highlighted with a blue checkmark. Below the query list is a 'Program Selection' section with a list of programs. The first two options are 'Genomic survey sequences (gsa)' and 'Sequence tagged sites (dsts)'. The third option is 'Sequence tagged sites (dsts)', which is highlighted with a blue checkmark. Below the program list is an 'Optimize for' section with a list of optimization options. The first two options are 'Sequence tagged sites (dsts)' and 'Sequence tagged sites (dsts)'. The third option is 'Sequence tagged sites (dsts)', which is highlighted with a blue checkmark.

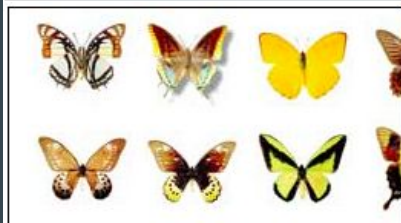
BLAST -- Basic Local Alignment Search Tool

- Sequence similarity search tool
- Finds related nucleotide and protein sequences
- Designed to find homologs
- Used for other sequence analysis tasks



BioProject

A BioProject is a collection of biological data related to a single initiative, originating from a single organization or from a consortium. A BioProject record provides users a single place to find links to the diverse data types generated for that project.



Taxonomy

The Taxonomy Database is a curated classification and nomenclature for all of the organisms in the public sequence databases. This currently represents about 10% of the described species of life on the planet.



Assembly

Genome assembly organization and additional information.



Gene

Gene integrates information from a wide range of species. A record may include nomenclature, Reference Sequences (RefSeqs), maps, pathways, variations, phenotypes, and links to genome-, phenotype-, and locus-specific resources worldwide.

Central Hubs: BioProject, Taxonomy, Genome, Assembly and, Gene

Entrez resources we will use today

Single-focus databases

- Nucleotide submitted and NCBI curated nucleotide sequences
- Protein coding regions translations of the above plus outside protein sources
- Sequence Read Archive (SRA) next-gen sequencing data

Information hubs (collection points)

- Taxonomy classification of organisms with data at NCBI
- BioProject collects data from a particular or project study
- Assembly data and information for assembled genome sequences and annotation
- Gene data and information related to specific genes

Sequence Data

Submitted nucleotide sequences and corresponding proteins

International Sequence Database Collaboration (INSDC)

- **Standard Sequences**
- **GenBank** -- US Sequence Database at NCBI
- **European Nucleotide Archive** at EBI
- **DNA Databank of Japan** at NIG

Next-Gen sequencing reads

- **Sequence Read Archive** (SRA)

High quality curated DNA and protein records

- **NCBI Reference Sequences**
 - **Swiss-Prot**
- **Proteins from PDB**

The Sequence Read Archive (SRA)

Repository for next-generation sequence data.

- Whole genome, metagenomic, and transcriptome data.
- Supporting data are in [BioProjects](#), [BioSample](#), [GEO datasets](#).
- Data [shared](#) with the other INSDC repositories
- Used in genome annotation pipelines and in Pathogen Project
- SRA Toolkit software for downloading / managing
essential for working with SRA data locally and as a client

www.ncbi.nlm.nih.gov/Traces/sra/

NCBI Reference Sequences (RefSeq)

- Produced and owned by NCBI from primary data
- Provide a reference standard
- Represent all molecules in the central dogma
- for selected eukaryotes: genomic, mRNA transcripts, proteins
- for prokaryotes and viruses: genomic and protein (no mRNA records)
- Distinct accession series – prefix and underscore “_”
- genomic: NC_, AC_, NG_, NZ_ (NC_000001, NZ_AAAM04000036)
- RNA: NM_, NR_, XM_, XR_ (NM_002020, XM_008782356)

RefSeq non-redundant proteins

Problem:

- Over 100 thousand RefSeq bacterial assemblies
- Many identical proteins in common

Solution:

- Make only one copy of a shared protein to annotate all

```
LOCUS      WP_004199234                293 aa          linear   BCT 01-FEB-2018
DEFINITION MULTISPECIES: carbapenem-hydrolyzing class A beta-lactamase KPC-2
            [Proteobacteria].
ACCESSION  WP_004199234
VERSION    WP_004199234.1
KEYWORDS   RefSeq.
SOURCE     Proteobacteria
            ORGANISM  Proteobacteria
                        Bacteria.
COMMENT    REFSEQ: This record represents a single, non-redundant, protein
            sequence which may be annotated on many different RefSeq genomes
            from the same, or different, species.
            COMPLETENESS: full length.
FEATURES   Location/Qualifiers
            source          1..293
                        /organism="Proteobacteria"
                        /db_xref="taxon:1224"
            Protein           1..293
                        /product="carbapenem-hydrolyzing class A beta-lactamase
                        KPC-2"
                        /EC_number="3.5.2.6"
                        /calculated_mol_wt=30984
            Region           4..264
                        /region_name="Transpeptidase"
                        /note="Penicillin binding protein transpeptidase domain;
                        c121491"
                        /db_xref="CDD:328749"
ORIGIN
1  mslyrrrlvll sclswplagf sataltnlva epfakleqdf ggsigvyamd tsgatvsyr
61  aerfplcss fkgflaaavl arsqqqagll dtpirygkna lvpwapisek ylttgmtvae
121  lsaaavqysd naaanllike lgspagltaf mrsigdttr ldrweleins aipgdardts
181  spravteslq kitlgsalaa pqrqqfvdwl kgnttgnhri raavpadwav gdkgtgcgvv
241  gtandyavvv ptgrapivla vytrapnkdd khseaviaaa arlaleglqv ngq
//
```

RefSeq non-redundant proteins

Problem:

- Over 100 thousand RefSeq bacterial assemblies
- Many identical proteins in common

Solution:

- Make only one copy of a shared protein to annotate all

ES: WP_004199234

<https://www.ncbi.nlm.nih.gov/sites/GQuery>

```
LOCUS      WP_004199234                293
DEFINITION MULTISPECIES: carbapenem-hydr
            [Proteobacteria].
ACCESSION  WP_004199234
VERSION    WP_004199234.1
KEYWORDS   RefSeq.
SOURCE     Proteobacteria
            ORGANISM  Proteobacteria
                        Bacteria.
COMMENT    REFSEQ: This record represent
            sequence which may be annotat
            from the same, or different,
            COMPLETENESS: full length.
FEATURES   Location/Qualifiers
            source     1..293
                        /organism="Proteobac
                        /db_xref="taxon:1224
            Protein    1..293
                        /product="carbapenem
                        KPC-2"
                        /EC_number="3.5.2.6"
                        /calculated_mol_wt=3
                        4..264
            Region     /region_name="Transp
                        /note="Penicillin bi
                        ci21491"
                        /db_xref="CDD:328749
ORIGIN
1 malyrrivll sclawplagf sataltnlv
61 aerfplcss fkgflaaavl arsqqqagl
121 lsaaavqysd naaanllike lgppaglt
181 spravteslq kitlgsalaa pqrqqfvdw
241 gtandyavvv ptgrapivla vytrapnkd
//
```

Taxonomic Groups

```
bacteria [204]
├── g-proteobacteria [203]
│   ├── enterobacteria [188]
│   │   ├── Enterobacteriaceae [173]
│   │   │   ├── Klebsiella [85]
│   │   │   ├── Enterobacter [41]
│   │   │   ├── Escherichia [15]
│   │   │   ├── Citrobacter [16]
│   │   │   ├── Lelercia [8]
│   │   │   ├── Raoultella [2]
│   │   │   ├── Pluralibacter [1]
│   │   │   └── Salmonella [2]
│   │   └── Yersiniaceae [6]
│   └── Morganellaceae [2]
│       ├── Proteus [1]
│       └── Morganella [1]
├── Erwiniaceae [6]
├── Phytobacter [1]
├── Pseudomonadales [5]
│   ├── Pseudomonadaceae [3]
│   └── Moraxellaceae [2]
├── Aeromonadales [10]
├── b-proteobacteria [1]
└── other sequences [2]
```

BioProject

Data by project

- Useful for multispecies projects
- RefSeq Targeted Loci project
- Metagenomes

Fossil & Extinct organisms

- Access to genome projects with no assembled data.
 - Companion database BioSample
- Metadata about data sources

- ☐ [fermentation metagenome](#)
 1. Pit mud of Chinese liquor fermentation reactorMetagenome
Taxonomy: [fermentation metagenome](#)
Project data type: **Metagenome**
Attributes : Scope: Environment; Material: Genome; Capture: Whole; Method Type: Sequencing
Chongqing university
Accession: PRJNA260163 ID: 260163
- ☐ [bioreactor sludge metagenome](#)
 2. Enhanced biological phosphorus removal bioreactor **Metagenome**
Taxonomy: [bioreactor sludge metagenome](#)
Project data type: **Metagenome**
Attributes : Scope: Environment; Material: Genome; Capture: Whole; Method Type: Sequencing
California Institute of Technology
Accession: PRJNA259882 ID: 259882
- ☐ [fermentation metagenome](#)
 3. Bacterial community
Taxonomy: [fermentation metagenome](#)
Project data type: **Metagenome**
Attributes : Scope: Environment; Material: Genome; Capture: Whole; Method Type: Sequencing
KAIST
Accession: PRJNA259685 ID: 259685
- ☐ [soil metagenome](#)
 4. China_Gurbantunggut Desert_Cyanobacteria **Metagenome**
Taxonomy: [soil metagenome](#)
Project data type: **Metagenome**
Attributes : Scope: Environment; Material: Genome; Capture: Whole; Method Type: Sequencing
Xinjiang Institute of Ecology and Geography
Accession: PRJNA259407 ID: 259407
- ☐ [human gut metagenome](#)
 5. Fecal microbiota of toddlers **Metagenome**
Taxonomy: [human gut metagenome](#)
Project data type: **Metagenome**
Attributes : Scope: Environment; Material: Genome; Capture: Whole; Method Type: Sequencing
The Ohio State University
Accession: PRJNA258387 ID: 258387

16S RefSeq Targeted Loci

BacteriaAccession: PRJNA33175 ID: 33175

Bacterial 16S Ribosomal RNA RefSeq Targeted Loci Project

The small subunit ribosomal RNA is a useful phylogenetic marker that has been used extensively for evolutionary analyses. The RefSeq dataset contains curated 16S ribosomal RNA sequences that correspond to bacteria type materials. The RefSeq records may contain corrections to the sequence or taxonomy as compared to the original INSD submission, and may have additional information added that is not found in the original. All sequences will have the same project ID and can be found as such.

[NCBI RefSeq Targeted Loci Project](#). [Less...](#)

NAVIGATE UP

This project is a component of the RefSeq Targeted Loci Project

NAVIGATE ACROSS

9 additional projects are components of the RefSeq Targeted Loci Project.

Accession	PRJNA33175
Data Type	RefSeq Targeted Locus (Loci)
Scope	Multisolate
Publications (total 10213)	1. Ma J <i>et al.</i> , "Rhodococcus gannanensis sp. nov., a novel endophytic actinobacterium isolated from root of sunflower (<i>Helianthus annuus</i> L.).", <i>Antonie Van Leeuwenhoek</i> , 2017 May 9;110(9):1113-1120 More...
Submission	Registration date: 10-Dec-2008 NCBI
NCBI Links	Targeted Loci Project
Related Resources	DSMZ Silva RDP GreenGenes StrainInfo CRW

Resource Name	Number of Links
SEQUENCE DATA	
Nucleotide (total)	19083
Genomic DNA	12
Transcript	19071
PUBLICATIONS	
PubMed	10213
PMC	1056

18/10/2018

Bacteria: <https://www.ncbi.nlm.nih.gov/bioproject/PRJNA33175>
Archaea: <https://www.ncbi.nlm.nih.gov/bioproject/PRJNA33317>

Pathogen AMR Reference Genes

Bacterial Antimicrobial Resistance Reference Gene Database		Accession: PRJNA313047 ID: 313047
This Bioproject contains annotated sequence records for representative DNA sequences that encode proteins conferring or contributing to resistance to various antibiotics. More...		
Accession	PRJNA313047	
Data Type	Targeted Locus (Loci)	
Scope	Multispecies	
Publications (total 1594)	1. Meunier D <i>et al.</i> , "FRI-2 carbapenemase-producing Enterobacter cloacae complex in the UK", <i>J Antimicrob Chemother</i> , 2017 Sep 1;72(9):2478-2482 More...	
Submission	Registration date: 24-Feb-2016 NCBI	
NCBI Links	- NCBI Pathogen Detection - National Database of Antibiotic Resistant Organisms More...	
Related Resources	- CARD - Lahey - Pasteur Institute Beta Lactamases - ResFinder - NCBI AMR FTP - NCBI AMR HMMs	
Relevance	Agricultural, Medical, Industrial, Environmental, Evolution	
Locus Tag Prefix	A7J11	

- Data part of Pathogen Detection Project
- RefSeq Nucleotide (NG_) and Protein (WP_) records
- Sequences and HMMs available on FTP

Resource Name	Number of Links
SEQUENCE DATA	
Nucleotide (Genomic DNA)	4331
Protein Sequences	4322
PUBLICATIONS	
PubMed	1594
PMC	965

BLAST Services and Databases

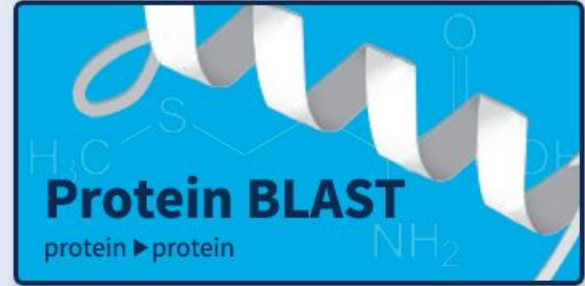
Web BLAST



Nucleotide BLAST
nucleotide ► nucleotide

blastx
translated nucleotide ► protein

tblastn
protein ► translated nucleotide



Protein BLAST
protein ► protein

BLAST Genomes

Enter organism common name, scientific name, or accession number

Human Mouse

Specialized searches

SmartBLAST

Find proteins highly similar to your query

Primer-BLAST

Design primers specific to your PCR template

Primer-BLAST

Design primers specific to your PCR template

Nucleotide BLAST Databases

Important databases for
blastn/tblastn
nr/nt

Traditional INSDC records plus

RefSeq RNA and
non-assembly RefSeqs 16S

RefSeq type strains PRJNA33175 and
PRJNA33317

Representative genomes
RefSeq reference and representative
assemblies

SRA

Next-gen reads by experiment

Standard Nucleotide BLAST

blastn [blastp](#) [blastx](#) [tblastn](#) [tblastx](#)

BLASTN programs search nucleotide databases using a nucleotide query. [more...](#)

Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [?](#)

NG_048935

[Clear](#)

Query subrange [?](#)

From

To

Or, upload file

[Choose File](#)

No file chosen [?](#)

Job Title

NG_048935:Escherichia coli blaCTX-M gene for...

Enter a descriptive title for your BLAST search [?](#)

☐ Align two or more sequences [?](#)

Choose Search Set

Database

☐ Human genomic + transcript ☐ Mouse genomic + transcript ☒ Others (nr etc.):

Nucleotide collection (nr/nt) [?](#)

Organism

[Optional](#)

enterobacteria (taxid:91347)

☐ Exclude [+](#)

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown [?](#)

Downloading Services / Protocols

- Entrez “Send/Send to” menu
- Entrez Assembly Download function
- WGS Browser www.ncbi.nlm.nih.gov/Traces/wgs/
- SRA www.ncbi.nlm.nih.gov/Traces/sra/
- SRA Toolkit www.ncbi.nlm.nih.gov/Traces/sra/?view=software
- Genomes/Assembly FTP areas
- GenBank [ftp.ncbi.nlm.nih.gov/genomes/all/](ftp://ftp.ncbi.nlm.nih.gov/genomes/all/)
- RefSeq [ftp.ncbi.nlm.nih.gov/genomes/refseq/](ftp://ftp.ncbi.nlm.nih.gov/genomes/refseq/)
- All [ftp.ncbi.nlm.nih.gov/genomes/all/](ftp://ftp.ncbi.nlm.nih.gov/genomes/all/)
- E-Utilities API / EDirect package

